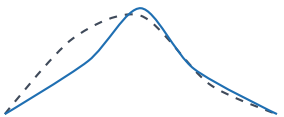
**1998**
**Alard & Lupton**
**OIS: kernel fitting**

**2008**
**Bramich**
**Discrete DIA**

**2016**
**ZOGY**
**Closed-form Fourier**

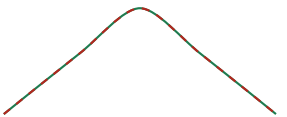
**2022**
**SFFT (Hu et al.)**
**Delta-basis space**

### 1998–2008: Kernel Fitting



Kernel estimated from brightest stars; vulnerable to crowding and nebular gas.

### 2016: ZOGY (closed form)



Closed-form Fourier division using known PSFs; no kernel ever fitted.

### Why ZOGY for SPHERExLens?

- **Analytic Null**: Significance $S \sim \mathcal{N}(0,1)$ where sky is constant.
- **Provable Threshold**: Gaussian tails yield exact false-alarm budgets ($2N^2\Phi(-\tau)$).
- **Speed**: Exactly two 2D-FFTs per epoch pair ($< 0.6\,\mathrm{s}$ per $180^2$ cutout).